

Task 2.1: Identification of DFD Elements

Processes, data stores, external entities and data flows of the BonVoyage system. Both DFD levels are generated from this single specification, which guarantees balancing.

Table 2.1(a): External entities

Entity	Role	Data supplied to the system	Data received from the system
Group Member	Registered participant who submits preferences and decides on packages and itineraries.	account credentials; preference profile & past trips; package & itinerary decisions	trip invitation; trip packages & cost breakdown; day-wise itinerary; notifications & trip history
Trip Organiser	Member who creates the trip, sets group parameters and selects a destination.	trip details & member list; destination selection	preference conflict report; ranked shortlist or diagnostic
System Administrator	Maintains the destination catalogue and fallback datasets.	catalogue & fallback data updates	catalogue status report
Maps & Routing Service	External provider of routes, distances and place reference data.	route & distance data; reference place data	route & distance query
Hotel & Fare Provider	External provider of accommodation options and indicative fares.	hotel & fare data	hotel & fare query
Language Model Service	External provider that composes itinerary text from grounded facts.	composed itinerary text	grounded itinerary prompt

Table 2.1(b): Processes (Level 1)

ID	Process	Responsibility
1.0	Manage Accounts & Trip Groups	Registers and authenticates users, creates trips and records membership.
2.0	Collect Preferences & Availability	Captures each member's structured preference profile and past trips.
3.0	Aggregate Group Profile	Builds the group model, computes the common window and detects conflicts.
4.0	Recommend Destinations	Filters by hard constraints, ranks by group-fit score and explains results.
5.0	Generate & Cost Packages	Assembles transport and stay options with an itemised cost breakdown.
6.0	Generate Itinerary	Composes a day-wise plan from verified facts for the accepted package.
7.0	Record Decisions & Notify	Stores accept, modify and reject decisions and issues notifications.
8.0	Maintain Destination Catalogue	Updates catalogue entries and curated fallback data.

Table 2.1(c): Data stores

ID	Data store	Contents	Written by	Read by
D1	User Accounts	User identity, credentials (salted hash) and home city.	1.0	1.0
D2	Trips & Memberships	Trip parameters, members, roles and invitation status.	1.0	2.0, 3.0
D3	Preference Profiles	Destination types, activities, budget range, availability, past trips.	2.0	3.0
D4	Destination Catalogue	Destinations, coordinates, cost bands and fallback fare data.	8.0	4.0, 5.0
D5	Packages & Cost Items	Generated packages, cost lines and decision status.	5.0, 7.0	6.0
D6	Itineraries	Day-wise itineraries, edits, decisions and trip history.	6.0, 7.0	7.0

Validation checklist (all rules satisfied)

✓

Every external flow in the Context Diagram appears at Level 1 with the same name and direction (balanced).

✓

No process is a black hole or a miracle: each has at least one input and one output.

✓

Every data store is written by at least one process and read by at least one process.

✓

No flow runs directly between two data stores or between an entity and a data store.

✓

Processes are named as verb phrases, data flows are named as noun phrases.

✓

The single External Service Provider actor of the use case diagram is realised by three service entities.

Table 2.1(d): External data flows (identical at Level 0 and Level 1)

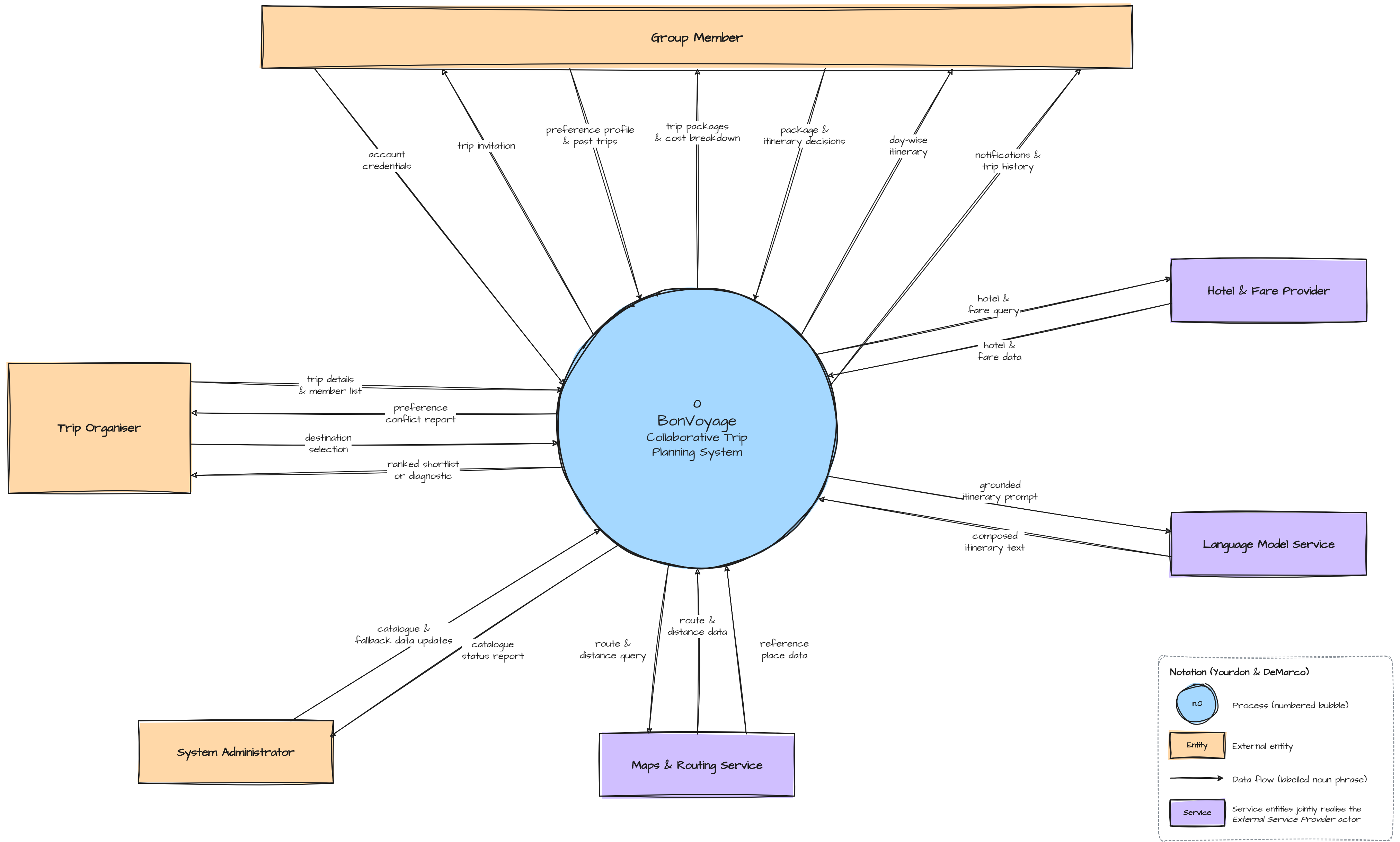
Data flow	Source	Destination
account credentials	Group Member	1.0 Manage Accounts & Trip Groups
trip invitation	1.0 Manage Accounts & Trip Groups	Group Member
preference profile & past trips	Group Member	2.0 Collect Preferences & Availability
trip packages & cost breakdown	5.0 Generate & Cost Packages	Group Member
day-wise itinerary	6.0 Generate Itinerary	Group Member
package & itinerary decisions	Group Member	7.0 Record Decisions & Notify
notifications & trip history	7.0 Record Decisions & Notify	Group Member
trip details & member list	Trip Organiser	1.0 Manage Accounts & Trip Groups
preference conflict report	3.0 Aggregate Group Profile	Trip Organiser
destination selection	Trip Organiser	4.0 Recommend Destinations
ranked shortlist or diagnostic	4.0 Recommend Destinations	Trip Organiser
catalogue & fallback data updates	System Administrator	8.0 Maintain Destination Catalogue
catalogue status report	8.0 Maintain Destination Catalogue	System Administrator
route & distance query	4.0 Recommend Destinations	Maps & Routing Service
route & distance data	Maps & Routing Service	4.0 Recommend Destinations
reference place data	Maps & Routing Service	8.0 Maintain Destination Catalogue
hotel & fare query	5.0 Generate & Cost Packages	Hotel & Fare Provider
hotel & fare data	Hotel & Fare Provider	5.0 Generate & Cost Packages
grounded itinerary prompt	6.0 Generate Itinerary	Language Model Service
composed itinerary text	Language Model Service	6.0 Generate Itinerary

Table 2.1(e): Internal data flows (Level 1)

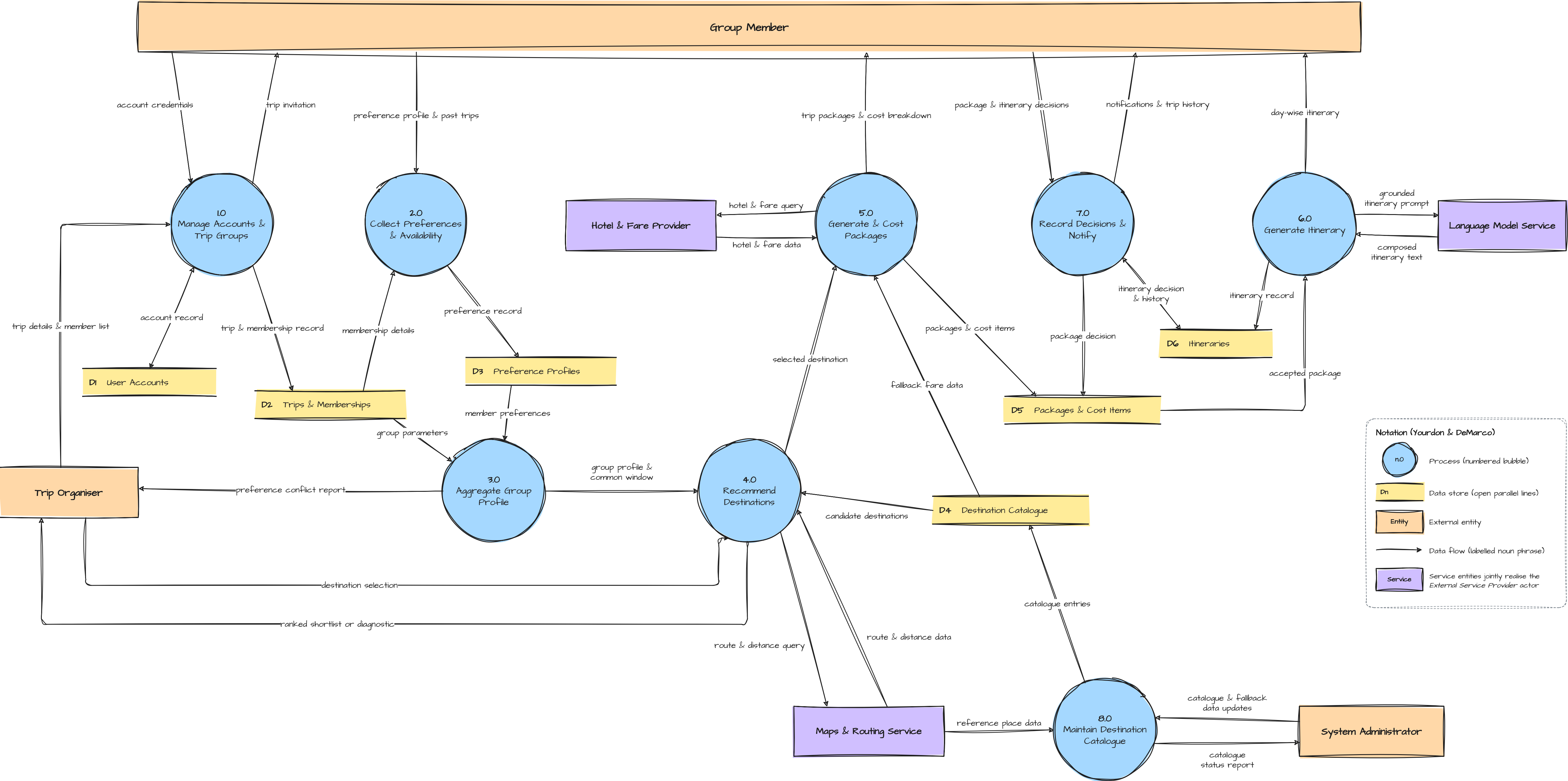
Data flow	Source	Destination
account record (read / write)	1.0 Manage Accounts & Trip Groups	D1 User Accounts
trip & membership record	1.0 Manage Accounts & Trip Groups	D2 Trips & Memberships
membership details	D2 Trips & Memberships	2.0 Collect Preferences & Availability
preference record	2.0 Collect Preferences & Availability	D3 Preference Profiles
group parameters	D2 Trips & Memberships	3.0 Aggregate Group Profile
member preferences	D3 Preference Profiles	3.0 Aggregate Group Profile
group profile & common window	3.0 Aggregate Group Profile	4.0 Recommend Destinations
candidate destinations	D4 Destination Catalogue	4.0 Recommend Destinations
selected destination	4.0 Recommend Destinations	5.0 Generate & Cost Packages
fallback fare data	D4 Destination Catalogue	5.0 Generate & Cost Packages
packages & cost items	5.0 Generate & Cost Packages	D5 Packages & Cost Items
package decision	7.0 Record Decisions & Notify	D5 Packages & Cost Items
accepted package	D5 Packages & Cost Items	6.0 Generate Itinerary
itinerary record	6.0 Generate Itinerary	D6 Itineraries
itinerary decision & history (read / write)	7.0 Record Decisions & Notify	D6 Itineraries
catalogue entries	8.0 Maintain Destination Catalogue	D4 Destination Catalogue

BonVoyage: Context Diagram (Level 0 DFD)

The complete system as a single process (O) and every external entity it exchanges data with.

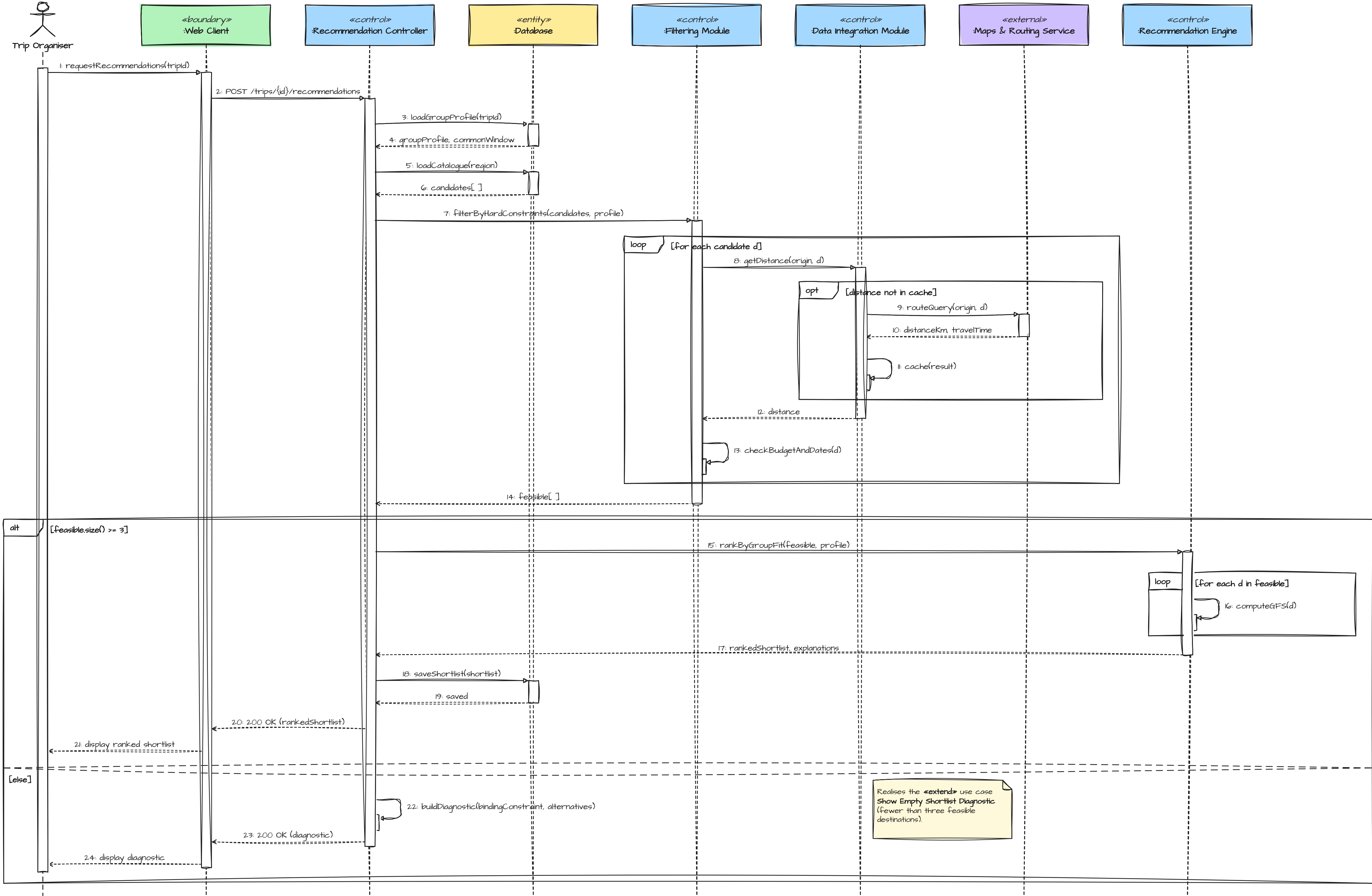


BonVoyage: Level 1 Data Flow Diagram
Decomposition of process 0 into eight major processes, six data stores and six external entities. Balanced against the Context Diagram (Level 0).



BonVoyage: Sequence Diagram for the use case Recommend Destinations

UML 2 notation. Includes Filter by Hard Constraints and Rank by Group-Fit Score; the else operand realises the extension Show Empty Shortlist Diagnostic.



Task 31: Selected use case

Recommend Destinations (primary actor: Trip Organiser). Chosen because it is the core of the system and exercises both kinds of use case relationship: it includes Filter by Hard Constraints and Rank by Group-Fit Score, and is extended by Show Empty Shortlist Diagnostic.

Task 32: Participating objects

Trip Organiser «actor»	Primary actor. initiates the use case and receives the shortlist or the diagnostic.
:Web Client «boundary»	Presentation layer. Captures the request and renders the result.
:Recommendation Controller «control»	Orchestrates the use case: loads data, calls the modules, decides between the two outcomes.
:Database «entity»	PostgreSQL store holding the group profile (D2, D3), the catalogue (D4) and saved shortlists.
:Filtering Module «control»	Geospatial and Filtering Module. Enforces the radius, budget, duration and date constraints.
:Data Integration Module «control»	Mediates all third-party calls; serves cached distances and caches new ones.
:Maps & Routing Service «external»	External system returning road distance and travel time.
:Recommendation Engine «control»	Computes the group-fit score for each feasible destination and builds explanations.

Notation

Solid line, filled head: synchronous call
Dashed line, open head: return message
White bar: execution specification (activation)
Frames: loop iteration, opt optional, alt alternatives, each with a guard in brackets
Participants are ordered left to right by first use.

Tasks 4.1 to 4.3: Entity, states and triggering events

Task 4.1: Chosen entity: Trip

A **Trip** is the object whose behaviour depends most on its history. The same event is legal in one state and meaningless in another: a member can accept a package only while packages are proposed, and *cancelTrip* is allowed throughout planning but only before the start date once the plan is finalised. Its lifecycle also traces the whole use case chain, from Manage Trip Group through Review and Decide on Itinerary to View Trip History, so the state machine doubles as a check that the use cases connect.

Task 4.2: States of Trip

#	State	Meaning	Entry / do behaviour
S1	Draft <i>(substate of Planning)</i>	Trip created by the organiser; parameters editable, no members yet.	entry / validateTripParameters
S2	Collecting Preferences <i>(substate of Planning)</i>	Invitations sent; members submit profiles, availability and past trips.	entry / openSubmissionWindow
S3	Ready for Recommendation <i>(substate of Planning)</i>	Every member has submitted; the group profile and common window are computed.	entry / computeGroupProfile
S4	Shortlisted <i>(substate of Planning)</i>	At least three feasible destinations ranked by group-fit score.	entry / saveRankedShortlist
S5	Packages Proposed <i>(substate of Planning)</i>	Costed packages for the chosen destination are under group review.	entry / generateAndCostPackages
S6	Package Selected <i>(substate of Planning)</i>	The group has accepted one package; its facts are frozen for itinerary generation.	entry / lockPackage
S7	Itinerary Proposed <i>(substate of Planning)</i>	A day-wise plan grounded in the package is under group review.	entry / assembleDayWisePlan
S8	Finalised	Plan confirmed by the group; waiting for the travel start date.	entry / notifyMembers
S9	In Progress	The trip is under way.	do / sendDailyReminders
S10	Completed	Trip ended; ratings requested and stored as past-trip data.	entry / archiveToTriptHistory
S11	Cancelled	Terminated by the organiser before completion.	entry / notifyMembers; releaseHolds
C	Planning <i>(composite)</i>	Encloses S1 to S7, so a single <i>cancelTrip</i> transition covers all seven planning substates.	

Task 4.3: Events that trigger transitions

#	From	Event	Guard	Action	To
T1	initial pseudostate	createTrip		assignOrganiser	Draft
T2	Draft	inviteMembers		sendInvitations	Collecting Preferences
T3	Collecting Preferences	preferencesSubmitted	[pending = 0]		Ready for Recommendation
T4	Ready for Recommendation	conflictDetected		reportConflict	Collecting Preferences
T5	Ready for Recommendation	preferenceRevised			Ready for Recommendation
T6	Ready for Recommendation	recommendationRequested		filterByHardConstraints	choice pseudostate
T7	choice pseudostate	<i>none (branch)</i>	[feasible >= 3]	rankByGroupFit	Shortlisted
T8	choice pseudostate	<i>none (branch)</i>	[else]	showEmptyShortlistDiagnostic	Collecting Preferences
T9	Shortlisted	destinationSelected			Packages Proposed
T10	Packages Proposed	packageRejected	[no alternative left]		Shortlisted
T11	Packages Proposed	packageAccepted			Package Selected
T12	Package Selected	generateItinerary			Itinerary Proposed
T13	Itinerary Proposed	itineraryRejected			Package Selected
T14	Itinerary Proposed	itineraryAccepted			Finalised
T15	Planning (any substate)	cancelTrip			Cancelled
T16	Finalised	cancelTrip	[now < startDate]		Cancelled
T17	Finalised	at(startDate)			In Progress
T18	In Progress	at(endDate)		requestRatings	Completed
T19	Completed	<i>completion</i>			final state
T20	Cancelled	<i>completion</i>			final state

Internal transitions (fire without exit or re-entry): *preferencesSubmitted* [pending > 0] / *updateProgress* in Collecting Preferences; *packageModified* / *recomputeCost* and *packageRejected* [alternatives left] / *hidePackage* in Packages Proposed; *itineraryModified* / *revalidate* in Itinerary Proposed. By contrast T5 is a true self-transition: re-entering Ready for Recommendation reruns its entry action, so revised preferences always rebuild the group profile.

BonVoyage: State Transition Diagram (UML State Machine) of the Trip entity

Transitions are labelled event [guard] / action. Lines inside a state are its entry or do behaviour and its internal transitions, which fire without leaving the state.

